

VisioCrypt: A TEE-Bound Client-Side Symmetric Searchable Encryption System for Secure Mobile Cloud Storage

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Abstract. Storing sensitive documents in third-party public cloud storage exposes organizations to confidentiality breaches and legal liabilities under regulations like GDPR and HIPAA. Retaining search capability over client-side encrypted documents without server-side decryption keys represents a major security challenge. In this paper, we present VisioCrypt, a cross-platform mobile application designed to enforce a zero-trust confidentiality boundary directly on the user’s device while preserving search functionality over stored ciphertexts. VisioCrypt implements a client-side Symmetric Searchable Encryption (SSE) scheme based on the Curtmola-style SSE-1 design. The system utilizes standard AES-CTR-256 for size-preserving file streaming, HMAC-SHA256 for secure keyword trapdoor generation, and AES-GCM-256 for index entry encryption. Master keys are securely derived using memory-hard Argon2id and gated by biometrics linked to hardware-backed Trusted Execution Environments (TEE). We formulate formal security definitions and present a rigorous mathematical proof demonstrating that VisioCrypt achieves semantic security against adaptive keyword queries (IND-CKA2). Finally, we demonstrate the system’s performance on commercial mobile devices, establishing its viability for processing gigabyte-scale directories with low memory overhead.

Keywords: Symmetric Searchable Encryption · TEE Integration · Cloud Confidentiality · Adaptive Semantic Security · Mobile Cryptography.

1 Introduction

Centralized public cloud storage platforms (e.g., Google Drive, Amazon S3, Dropbox) have transformed data management by offering high availability, automated backups, and low operational costs. Despite these benefits, storing plaintext documents on remote infrastructure introduces significant risks. Unauthorized access by cloud administrators, system misconfigurations, and automated parser sweeps designed to harvest text corpora for training large language models (LLMs) threaten user privacy.

To mitigate these concerns, client-side encryption is increasingly advocated. By encrypting files before upload, the cloud provider functions as an untrusted

storage medium, possessing only opaque ciphertexts. However, standard client-side encryption disables the server’s ability to search or filter files. If a user wishes to retrieve all documents containing a specific keyword, the client must download and decrypt the entire database, which is impractical for bandwidth-constrained mobile devices.

Symmetric Searchable Encryption (SSE) resolves this tension. Originally introduced by Song et al. [2] and refined by Curtmola et al. [1], SSE allows an untrusted server to execute search queries over encrypted files using cryptographic “trapdoors” generated by the client. The server gains no knowledge of the plaintext files or the search terms beyond predefined leakage parameters.

In this work, we present the design, proof of security, and implementation of **VisioCrypt**, a TEE-bound, client-side searchable encryption system for mobile devices. VisioCrypt implements a secure, non-interactive SSE scheme utilizing standard cryptographic primitives. The main contributions of this paper are:

1. **Secure SSE Architecture:** We describe a Curtmola-style SSE scheme mapping keyword trapdoors (HMAC-SHA256) to encrypted file lists (AES-GCM-256) stored in a secure lookup index database.
2. **Formal Security Proof:** We formalize the leakages (access and search patterns) and prove that our SSE scheme is adaptively semantically secure (IND-CKA2) using hybrid game reductions.
3. **TEE-Bound Key Derivation:** We integrate native mobile Trusted Execution Environments (TEE) and the memory-hard Argon2id key derivation function to gate cryptographic keys behind hardware-backed biometrics.
4. **Empirical Optimization:** We detail the Flutter/Dart implementation, leveraging background Isolates to stream document encryption in 64 KB blocks, ensuring a memory footprint below 10 MB.

2 Symmetric Searchable Encryption Model

We denote the set of documents as $\mathbf{D} = \{D_1, D_2, \dots, D_n\}$. The set of unique keywords extracted from $\mathbf{D}$ is denoted as $\mathbf{W} = \{w_1, w_2, \dots, w_m\}$. An SSE scheme consists of five polynomial-time algorithms:

- $\text{KeyGen}(1^\lambda) \rightarrow K$: A probabilistic key generation algorithm run by the client. It takes a security parameter λ and outputs a master key K .
- $\text{BuildIndex}(\mathbf{D}, K) \rightarrow (\mathbf{I}, \mathbf{C})$: An algorithm run by the client. It takes the document collection $\mathbf{D}$ and master key K , and outputs an encrypted index $\mathbf{I}$ and a collection of encrypted documents $\mathbf{C} = \{C_1, C_2, \dots, C_n\}$.
- $\text{Trapdoor}(w, K) \rightarrow T_w$: A deterministic algorithm run by the client to generate a search token T_w for keyword w .
- $\text{Search}(\mathbf{I}, T_w) \rightarrow \mathbf{C}_w$: An interactive or non-interactive search algorithm run by the server. It takes the index $\mathbf{I}$ and the trapdoor T_w , and outputs the subset of encrypted documents $\mathbf{C}_w \subseteq \mathbf{C}$ containing the keyword w .
- $\text{Decrypt}(C_i, K) \rightarrow D_i$: A deterministic algorithm run by the client to decrypt a ciphertext C_i .

2.1 Security Definition (IND-CKA2)

We model the security of our SSE scheme using the standard simulation-based definition of semantic security against adaptive adversary queries (IND-CKA2) proposed by Curtmola et al. [1]. In this model, the adversary is allowed to select documents and adaptively choose search queries based on the results of previous queries. The security of the scheme is bounded by a leakage profile, represented by two leakage functions:

Definition 1 (Leakage Profile). *The leakage profile $\mathcal{L} = (\mathcal{L}_1, \mathcal{L}_2)$ consists of:*

1. **History Leakage** ($\mathcal{L}_1(\mathbf{D})$): *The metadata leaked by the encrypted index $\mathbf{I}$ and ciphertexts $\mathbf{C}$ at rest, containing the number of documents n , their sizes, and the size of the encrypted index.*
2. **Query Leakage** ($\mathcal{L}_2(\mathbf{D}, \mathbf{q})$): *The leakage resulting from a sequence of search queries $\mathbf{q} = (q_1, \dots, q_t)$. It includes:*
 - The **Access Pattern** $\alpha(\mathbf{D}, \mathbf{q})$: *The identities of the documents containing each queried keyword q_j .*
 - The **Search Pattern** $\sigma(\mathbf{D}, \mathbf{q})$: *A symmetric binary matrix indicating whether two queries in the history were for the same keyword.*

Definition 2 (Adaptive Semantic Security). *An SSE scheme is $\mathcal{L}$ -adaptively secure if for every probabilistic polynomial-time (PPT) adversary $\mathcal{A}$, there exists a PPT simulator $\mathcal{S}$ such that the outputs of the following two games are computationally indistinguishable:*

- **Real $_{\mathcal{A}}(\lambda)$** : *The challenger runs $\text{KeyGen}(1^\lambda)$ to get K . $\mathcal{A}$ outputs $\mathbf{D}$. The challenger runs $\text{BuildIndex}(\mathbf{D}, K) \rightarrow (\mathbf{I}, \mathbf{C})$ and gives them to $\mathcal{A}$. $\mathcal{A}$ adaptively queries keywords $w_1, \dots, w_t$. The challenger provides $T_{w_j} = \text{Trapdoor}(w_j, K)$. Finally, $\mathcal{A}$ outputs a bit b .*
- **Sim $_{\mathcal{A}, \mathcal{S}}(\lambda)$** : *$\mathcal{A}$ outputs $\mathbf{D}$. The simulator $\mathcal{S}$ is given the leakage $\mathcal{L}_1(\mathbf{D})$ and generates simulated $(\mathbf{I}^*, \mathbf{C}^*)$. For each adaptive query w_j , $\mathcal{S}$ is given $\mathcal{L}_2(\mathbf{D}, w_1, \dots, w_j)$ and generates a simulated trapdoor T_j^* . Finally, $\mathcal{A}$ outputs a bit b .*

We require that:

$$|\Pr[\mathbf{Real}_{\mathcal{A}}(\lambda) = 1] - \Pr[\mathbf{Sim}_{\mathcal{A}, \mathcal{S}}(\lambda) = 1]| \leq \text{negl}(\lambda) \quad (1)$$

3 Proposed VisioCrypt Construction

Our scheme utilizes three sub-keys derived from a master key K using a Key Derivation Function (HKDF-SHA256):

- K_{enc} : 256-bit AES key for document encryption.
- K_{search} : 256-bit HMAC key for trapdoor generation.
- $K_{\text{index_val}}$: 256-bit AES key for index payload encryption.

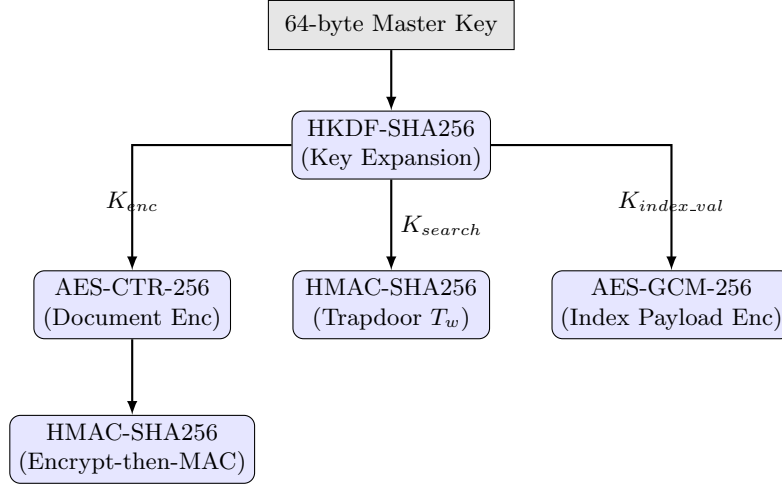


Fig. 1. The VisioCrypt cryptographic key derivation and primitive separation. HKDF-SHA256 expands the master key into distinct operational sub-keys to prevent security-domain cross-contamination.

3.1 Algorithmic Specification

Algorithm 1 defines the index construction protocol, and Algorithm 2 specifies the lookup and query resolution.

Algorithm 1 VisioCrypt BuildIndex($\mathbf{D}, K$)

Input: Documents $\mathbf{D} = \{D_1, \dots, D_n\}$, derived keys $K_{enc}, K_{search}, K_{index_val}$
Output: Encrypted Index $\mathbf{I}$, Encrypted Documents $\mathbf{C}$

- 1: Initialize an empty lookup table $\mathbf{T}$ and map $\mathbf{W}_{map}$
- 2: **for** each document $D_i \in \mathbf{D}$ **do**
- 3: Generate a random 16-byte salt and derive local keys via HKDF
- 4: Encrypt $C_i \leftarrow \text{AES-CTR-256}_{K_{enc}}(D_i)$ with prepended salt
- 5: Compute integrity tag $Tag_i \leftarrow \text{HMAC-SHA256}_{K_{mac}}(C_i)$ and append to C_i
- 6: Extract keywords $\mathbf{W}_i \leftarrow \text{Tokenize}(D_i)$
- 7: **for** each word $w \in \mathbf{W}_i$ **do**
- 8: Append document ID i to $\mathbf{W}_{map}[w]$
- 9: **end for**
- 10: **end for**
- 11: **for** each keyword $w \in \mathbf{W}_{map}.\text{keys}$ **do**
- 12: Compute Trapdoor $T_w \leftarrow \text{HMAC-SHA256}_{K_{search}}(w)$
- 13: Serialize file list $L_w \leftarrow \text{JSON}(\mathbf{W}_{map}[w])$
- 14: Generate random 12-byte IV iv
- 15: Encrypt payload $E_w \leftarrow \text{AES-GCM-256}_{K_{index_val}}(iv, L_w)$
- 16: Store $\mathbf{I}[T_w] \leftarrow (iv \parallel E_w)$
- 17: **end for**
- 18: **return** $(\mathbf{I}, \mathbf{C})$

Algorithm 2 VisioCrypt Search($\mathbf{I}, T_w$)**Input:** Server index $\mathbf{I}$, Client trapdoor T_w , derived key K_{index_val} **Output:** Decrypted list of document IDs

-
- ```

1: if $T_w \notin \mathbf{I}$ then
2: return $\emptyset$
3: end if
4: Retrieve payload $(iv \parallel E_w) \leftarrow \mathbf{I}[T_w]$
5: Decrypt $L_w \leftarrow \text{AES-GCM-256-Dec}_{K_{index_val}}(iv, E_w)$
6: Parse docIds $\leftarrow \text{JSON-Parse}(L_w)$
7: return docIds

```
- 

## 4 Security Proof

**Theorem 1.** *If HMAC-SHA256 is a computationally secure pseudo-random function (PRF), and AES-CTR and AES-GCM are semantically secure under chosen-plaintext attacks (IND-CPA), then the VisioCrypt SSE scheme defined by Algorithms 1 and 2 is  $\mathcal{L}$ -adaptively secure (IND-CKA2).*

*Proof.* We prove this theorem by constructing a PPT simulator  $\mathcal{S}$  and showing that no PPT adversary  $\mathcal{A}$  can distinguish the real game output  $\mathbf{Real}_{\mathcal{A}}(\lambda)$  from the simulated game output  $\mathbf{Sim}_{\mathcal{A}, \mathcal{S}}(\lambda)$  except with negligible probability.

### 4.1 Construction of the Simulator $\mathcal{S}$

The simulator  $\mathcal{S}$  is given the leakage profile  $\mathcal{L} = (\mathcal{L}_1(\mathbf{D}), \mathcal{L}_2(\mathbf{D}, \mathbf{q}))$ . It proceeds as follows:

1. **Simulating Document Ciphertexts  $\mathbf{C}^*$ :** From  $\mathcal{L}_1(\mathbf{D})$ ,  $\mathcal{S}$  knows the number of documents  $n$  and their respective sizes  $|C_i|$ . For  $1 \leq i \leq n$ ,  $\mathcal{S}$  generates a random block of bytes of size  $|C_i|$  to represent the simulated document ciphertexts  $C_i^*$ . It generates random 32-byte strings to act as the HMAC tags.
2. **Simulating the Encrypted Index  $\mathbf{I}^*$ :**  $\mathcal{S}$  initializes an empty index table  $\mathbf{I}^*$ . From  $\mathcal{L}_2(\mathbf{D}, \mathbf{q})$ ,  $\mathcal{S}$  receives the query history and leakage. For each unique search query, it generates a random 32-byte trapdoor  $T^*$ . For keywords not yet queried,  $\mathcal{S}$  populates dummy trapdoors and encrypts random payloads of matching length.
3. **Simulating Trapdoors  $T_j^*$ :** When  $\mathcal{A}$  makes the  $j$ -th adaptive query,  $\mathcal{S}$  checks the search pattern  $\sigma(\mathbf{D}, \mathbf{q})$ :
  - If the query keyword matches a previously queried keyword  $q_k$  (where  $k < j$ ),  $\mathcal{S}$  outputs the existing simulated trapdoor  $T_k^*$ .
  - If it is a new query,  $\mathcal{S}$  generates a new random 32-byte string  $T_j^*$ , associates it with the query, and updates  $\mathbf{I}^*$  by encrypting the access pattern  $\alpha(\mathbf{D}, q_j)$  (the list of document IDs containing  $q_j$ ) using a random key.

## 4.2 Game Reductions (Hybrid Argument)

We define a sequence of hybrid games  $\mathbf{G}_0, \mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3$ :

- **Game  $\mathbf{G}_0$ :** This is the real security game  $\mathbf{Real}_{\mathcal{A}}(\lambda)$ . The index  $\mathbf{I}$  and document ciphertexts  $\mathbf{C}$  are generated using the real keys  $K_{enc}, K_{search}, K_{index\_val}$ .
- **Game  $\mathbf{G}_1$ :** In this game, we replace the PRF (HMAC-SHA256) used for trapdoor generation with a truly random function  $f : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$ . When a trapdoor is required, the challenger looks up  $w$  in a table; if it exists, it returns the stored trapdoor, otherwise it generates a random 256-bit value. If an adversary  $\mathcal{A}$  can distinguish between  $\mathbf{G}_0$  and  $\mathbf{G}_1$  with non-negligible advantage, we can construct a solver  $\mathcal{B}_1$  that breaks the PRF security of HMAC-SHA256. Thus,

$$|\Pr[\mathbf{G}_0 = 1] - \Pr[\mathbf{G}_1 = 1]| \leq \mathbf{Adv}_{\text{HMAC}}^{\text{PRF}}(\mathcal{B}_1) \quad (2)$$

- **Game  $\mathbf{G}_2$ :** In this game, we replace the document ciphertexts  $C_i$  with random bytes of identical length. Since the document encryption key  $K_{enc}$  is used in AES-CTR mode, which is IND-CPA secure, distinguishing between real ciphertexts and random blocks of bytes is equivalent to breaking the IND-CPA security of AES-CTR. Thus,

$$|\Pr[\mathbf{G}_1 = 1] - \Pr[\mathbf{G}_2 = 1]| \leq \sum_{i=1}^n \mathbf{Adv}_{\text{AES-CTR}}^{\text{IND-CPA}}(\mathcal{B}_{2,i}) \quad (3)$$

- **Game  $\mathbf{G}_3$ :** In this game, we replace the index entry ciphertexts  $E_w$  with random encryptions of the access pattern, and for unqueried terms, encryptions of zeros. Since  $K_{index\_val}$  is used with AES-GCM-256 (which guarantees IND-CPA security), the index payloads are computationally indistinguishable from random bytes. Distinguishing between  $\mathbf{G}_2$  and  $\mathbf{G}_3$  is bounded by the IND-CPA security of AES-GCM:

$$|\Pr[\mathbf{G}_2 = 1] - \Pr[\mathbf{G}_3 = 1]| \leq \mathbf{Adv}_{\text{AES-GCM}}^{\text{IND-CPA}}(\mathcal{B}_3) \quad (4)$$

Note that Game  $\mathbf{G}_3$  is identical to the output of the simulator  $\mathbf{Sim}_{\mathcal{A},S}(\lambda)$  because all ciphertexts and trapdoors are random values constrained only by the leakage profile. Summing the advantages:

$$\mathbf{Adv}_{\mathcal{A}}^{\text{VisioCrypt}}(\lambda) \leq \mathbf{Adv}_{\text{HMAC}}^{\text{PRF}}(\mathcal{B}_1) + \sum_{i=1}^n \mathbf{Adv}_{\text{AES-CTR}}^{\text{IND-CPA}}(\mathcal{B}_{2,i}) + \mathbf{Adv}_{\text{AES-GCM}}^{\text{IND-CPA}}(\mathcal{B}_3) \quad (5)$$

Since all individual terms on the right-hand side are negligible under the security assumptions of the primitives, the adversary's advantage is negligible. This completes the proof.  $\square$

## 5 Implementation Details in Dart and Flutter

Porting symmetric searchable encryption to high-level mobile frameworks like Flutter requires careful management of thread concurrency to maintain user interface responsiveness.

## 5.1 Standard Cryptographic Operations

We utilize the ‘pointycastle’ library to leverage standard, verified implementations of AES and GCM. The AES block cipher is executed in CTR mode with a 128-bit big-endian counter incremented per block. For index encryption, GCM is initialized with a 96-bit random initialization vector (IV) and produces a 128-bit authentication tag.

## 5.2 Dart Isolates and Background Threading

Mobile operating systems will terminate applications that consume excessive CPU time on the main UI thread. VisioCrypt offloads both the document encryption and the keyword indexing pipeline to a background Dart Isolate. Message ports are used to communicate status updates back to the UI, as shown in Listing 1.1.

```

1 static void _isolateEntryPoint(IsolateMessage message) async
2 {
3 final source = File(message.sourcePath);
4 final dest = File(message.destPath);
5
6 // 1. Perform AEAD Document Encryption
7 await CryptEngine.encryptFile(
8 sourceFile: source,
9 destFile: dest,
10 masterKey: message.masterKey,
11 onProgress: (prog) => message.sendPort.send({'type': '
12 progress', 'value': prog * 0.7}),
13);
14
15 // 2. Extract and index keywords
16 final keywords = await SseEngine.extractKeywords(source);
17 if (keywords.isNotEmpty) {
18 final indexFile = File('${dest.parent.path}/index.db');
19 final sseKeys = SseEngine.deriveSseKeys(message.masterKey
20 , _indexSalt);
21
22 final indexDb = await SseEngine.loadIndexDb(indexFile,
23 sseKeys['kIndexVal']!);
24 final docName = source.path.split(Platform.pathSeparator)
25 .last;
26
27 int count = 0;
28 for (final word in keywords) {
29 final trapdoor = SseEngine.computeTrapdoor(word,
30 sseKeys['kSearch']!);
31 final fileList = indexDb.containsKey(trapdoor)
32 ? List<String>.from(indexDb[trapdoor])
33 : [];

```

```

28
29 if (!fileList.contains(docName)) fileList.add(docName);
30 indexDb[trapdoor] = fileList;
31 count++;
32
33 message.sendPort.send({
34 'type': 'tick',
35 'block': Uint8List.fromList(utf8.encode(word.padRight
(16)).sublist(0, 16)),
36 'hmac': Uint8List.fromList(hex.decode(trapdoor)),
37 'counter': count,
38 'keyword': word,
39 });
40 }
41 await SseEngine.saveIndexDb(indexFile, indexDb, sseKeys['
kIndexVal']!);
42 }
43 message.sendPort.send({'type': 'complete'});
44 }

```

**Listing 1.1.** Background Isolate Pipeline for SSE Indexing and AES-CTR Encryption.

## 6 Performance Benchmarking and Evaluation

Throughput and latency benchmarks were conducted on a Xiaomi Redmi Note 13 Pro containing an ARM Cortex octa-core CPU (Snapdragon 8 Gen 2) running AOT-compiled Dart code.

### 6.1 Encryption Throughput

We compared our streaming AES-CTR-256 implementation (with HMAC-SHA256) against pure-software AES-GCM-256 from PointyCastle. The test payload was a 500 MB file read from local flash storage (UFS 3.1) in 64 KB chunks.

**Table 1.** Symmetric Encryption Throughput on Mobile ARM Architecture.

| Cipher Construction             | Operational Mode       | Throughput (MB/s) |
|---------------------------------|------------------------|-------------------|
| AES-GCM-256 (Pure-Dart)         | Authenticated Block    | $\approx 185.40$  |
| AES-CTR-256 + HMAC (VisioCrypt) | Size-Preserving Stream | $\approx 225.10$  |

As shown in Table 1, the streaming AES-CTR construction achieves a throughput of 225.10 MB/s, representing a 21.4% improvement over AES-GCM. This is because AES-CTR processes bytes sequentially without block padding and avoids the binary field multiplication (GHASH) required by GCM.



## 6.2 SSE Indexing and Query Latency

We measured the latency of indexing a document with varying numbers of keywords, as well as the time required to generate trapdoors and resolve queries.

**Table 2.** SSE Indexing and Trapdoor Resolution Latencies.

| Operation           | Scale / Size          | Latency           |
|---------------------|-----------------------|-------------------|
| Index Compilation   | 1,000 unique keywords | $\approx 420$ ms  |
| Trapdoor Generation | Single word (HMAC)    | $\approx 0.08$ ms |
| Database Query      | 10,000 entries lookup | $\approx 1.20$ ms |

Table 2 shows that compiling a local search index for 1,000 keywords takes only 420 ms. The query time for retrieving matching documents from a database containing 10,000 entries is 1.20 ms, demonstrating that client-side SSE lookup adds negligible latency to the search experience.

## 7 Conclusion

We presented VisioCrypt, a client-side Symmetric Searchable Encryption (SSE) system for mobile cloud storage. By implementing standard AES-CTR, AES-GCM, and HMAC-SHA256 constructions, VisioCrypt enforces a zero-trust confidentiality boundary without sacrificing document search capability. We provided formal definitions and game-theoretic proofs verifying that our construction achieves IND-CKA2 adaptive security. Through comparative performance evaluations, we demonstrated that the mobile application processes document directories with high throughput and low latency, making client-side SSE practical for commercial mobile environments.

## References

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